Armin Salmasi

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WORK EXPERIENCE

Software Engineer & Model Developer *2023 - Present*
Thermo-Calc Software AB

- Designed and developed scalable Python backend services, APIs, and ML models for material property predictions in a production environment.
- Built and maintained internal tools and platforms to streamline data workflows and ML model deployments for cross-functional engineering teams.
- Engineered agentic AI workflows and multi-agent orchestration systems leveraging LLMs, prompt engineering, and tool-use agents.
- Managed CI/CD pipelines, automated testing, and build environments, ensuring high reliability for production systems.

Key Projects:

- **GenAI Toolchain:** Engineered agentic GenAI systems and code assistants using RAG, LLMs, and tool-use orchestration to automate development tasks.
- **ML Platform & Workflow:** Spearheaded end-to-end development and orchestration of ML workloads, enabling self-service and automated execution of complex computational tasks.
- **Data Infrastructure:** Developed robust ETL pipelines and deployed scalable NoSQL database solutions for efficient storage and retrieval of scientific data.

R&D Simulation Specialist *2023*
Epiroc Drilling Tools AB

- Analyzed large-scale manufacturing datasets to optimize production workflows and build data-driven design capabilities.
- Collaborated with cross-functional teams to translate domain-specific challenges into scalable computational and statistical models.

Key Projects:

- Data-driven design optimization using advanced statistical analysis and predictive modeling.
- Simulation of complex physical dynamics under high-pressure conditions.

Research Intern (Master Thesis) *2016*
Sandvik Coromant AB

- Applied **Density Functional Theory (DFT)** calculations to screen high-entropy alloy candidates.
- Developed diffusion simulation models to predict gradient formation during sintering.

TECHNICAL SKILLS

Languages	Python (Advanced), Java, Fortran, SQL, Bash Script
Platform & DevOps	GCP, Docker, Kubernetes, GitLab CI/CD, GitHub Actions, Jenkins, Git, ETL
ML & AI	PyTorch, TensorFlow, MLFlow, LLMs (RAG, Prompt Eng.), LangChain, Scikit-learn
Data & Analytics	Data Analysis, Descriptive & Inferential Statistics, Hypothesis Testing, SciPy (Stats), Pandas, NumPy, MongoDB, MySQL, Seaborn, Matplotlib, Plotly
HPC & Distributed	MPI (Fortran), OpenMP (Fortran), mpi4py, SLURM, Dask, PySpark, Celery, Multiprocessing, Threading, Asyncio, Concurrent
OS	Linux, macOS, Windows
IDEs & Copilots	IntelliJ, PyCharm, VSCode, Cursor, Antigravity, Claude Code, Codex

CERTIFICATIONS

- **Applied AI & Data Science Program**, MIT (Ongoing, Batch 2026)
- **Supervised ML: Regression & Classification**, DeepLearning.AI
- **Improving Deep Neural Networks**, DeepLearning.AI
- **Neural Networks & Deep Learning**, DeepLearning.AI
- **Structuring Machine Learning Projects**, DeepLearning.AI
- **Practical Deep Learning (HPC)**, ENCCS
- **CodeRefinery (Software Development)**, NeIC
- **Software Tools: ML to Phase Diagrams**, MGF
- **CSC HPC Summer School**, Solvalla, Finland
- **PDC HPC Summer School**, KTH, Sweden

EDUCATION

Ph.D., Engineering Materials Science
KTH Royal Institute of Technology
Focus: ICME, Mass Transport Simulation, and Numerical Modeling. Served as a Teaching Assistant for multiple advanced master's level and PhD courses.

M.Sc., Materials Science
KTH Royal Institute of Technology
Focus: Simulation of Gradient Formation using Thermodynamic and Kinetic modeling.

PUBLICATIONS

Please refer to my **Google Scholar Profile** for a complete and up-to-date list of my publications.